From RF Signals to Disease Diagnosis: A Pipeline-Centric Review of Wearable Radio Frequency Sensing for Multi-Disease Health Monitoring (2023–2025)

Deepa Asthana* and Praveen Asthana† *Department of Electronics and Communication Engineering
†Independent Researcher
Email: temp.genai18@gmail.com

***Abstract*—Wearable radio frequency (RF) sensing has emerged as a transformative paradigm for continuous, non-invasive health monitoring, yet existing reviews treat individual RF technologies in isolation without exposing the shared signal-processing pipeline that underpins all of them. This paper presents the first unified, end-to-end processing pipeline review spanning four major RF modalities—Frequency-Modulated Continuous Wave (FMCW) radar, Ultra-Wideband (UWB) impulse radio, millimeter-wave (mmWave) sensing, and microwave resonator/antenna-based systems—covering literature from 2023 to 2025. We decompose the pipeline into five canonical stages: (i) RF signal acquisition, (ii) data preparation and preprocessing, (iii) feature extraction and representation, (iv) machine learning and deep learning classification, and (v) multi-disease diagnostic output. Particular emphasis is placed on the preprocessing stage, where we provide the most detailed cross-technology comparison of clutter removal, motion artifact compensation, signal transformation, vital sign extraction, and data augmentation techniques reported to date. Across 30+ recent studies, we benchmark pipeline configurations for cardiac, respiratory, glucose, oncological, neurological, and sleep disorders, identifying that hybrid CNN-LSTM architectures operating on range-Doppler features from FMCW radar achieve the highest mean accuracy (96.2±1.8%) for cardiac monitoring, while UWB with wavelet-based preprocessing leads respiratory assessment (94.7±2.1%). We conclude with per-stage challenge analysis and a roadmap toward standardized, multi-modal RF health sensing frameworks.**

***Index Terms*—Wearable sensors, radio frequency sensing, FMCW radar, ultra-wideband, millimeter-wave, microwave sensing, signal processing pipeline, machine learning, deep learning, health monitoring, disease diagnosis.**

I. Introduction

CHRONIC diseases account for over 74% of global mortality, with cardiovascular disorders, diabetes, respiratory conditions, and cancer collectively responsible for an estimated 41 million deaths annually [1]. Traditional clinical monitoring—electrocardiography, polysomnography, spirometry—demands hospital visits, restricts patient mobility, and captures only episodic snapshots of physiological state. The unmet need for continuous, unobtrusive, and non-invasive monitoring has catalyzed rapid innovation in wearable sensing technologies [2].

Radio frequency (RF) sensing offers a compelling solution. Unlike optical or electrochemical sensors that require skin contact, RF signals penetrate clothing and tissue, enabling contactless or minimally intrusive measurement of vital signs, tissue dielectric properties, and physiological motion [3]. Four RF modalities have reached sufficient maturity for wearable health applications: Frequency-Modulated Continuous Wave (FMCW) radar, Ultra-Wideband (UWB) impulse radio, millimeter-wave (mmWave) phased arrays, and microwave resonator/antenna systems. Each produces fundamentally different raw signals—chirp beat frequencies, channel impulse responses, beamformed spatial spectra, and scattering parameters, respectively—yet all must traverse an analogous processing pipeline before yielding clinically actionable diagnoses.

Gap. Existing reviews either focus on a single RF technology [4], [5] or survey machine learning models without detailing the preprocessing stages that critically determine downstream accuracy [6]. No prior work presents a unified, stage-by-stage pipeline comparison across all four RF modalities for multi-disease monitoring.

Research Questions. This review addresses: **RQ1**—What are the optimal signal acquisition configurations per RF modality for wearable health sensing? **RQ2**—Which preprocessing and artifact removal techniques yield the greatest improvement per technology? **RQ3**—How do feature representations and classification models compare across the pipeline? **RQ4**—What pipeline configuration maximizes diagnostic accuracy for each target disease?

Contributions. (1) First pipeline-centric cross-technology review of RF wearable health sensing (2023–2025). (2) Deepest analysis of preprocessing techniques across FMCW, UWB, mmWave, and microwave modalities. (3) Comprehensive benchmarking of pipeline configurations mapped to six disease categories. (4) Identification of per-stage challenges and a roadmap for standardized RF health frameworks.

II. RF Signal Acquisition Layer

The acquisition layer defines the physical interface between the RF transceiver and the human body. We characterize four modalities by their signal generation, propagation, and digitization mechanisms.

A. FMCW Radar

FMCW radar transmits a linearly swept chirp signal $s_{tx}(t) = A\cos(2\pi f_0 t + \pi \frac{B}{T} t^2)$ where B is bandwidth and

Fig. 1. Unified end-to-end processing pipeline for wearable RF health sensing. All four RF modalities share five canonical stages, though the specific algorithms differ per technology. This paper reviews each stage in depth, with particular focus on the preprocessing layer (Section III).

T is chirp duration. The reflected signal is mixed with the transmit reference to produce an intermediate frequency (IF) signal whose frequency is proportional to target range [7]. Wearable FMCW systems operating at 60–77 GHz achieve sub-millimeter range resolution ($\Delta R = c/2B$), enabling chest wall displacement measurement for cardiac and respiratory monitoring [8]. The IF signal is digitized at 2–10 MSPS via on-chip ADCs in devices such as the Texas Instruments IWR1443BOOST and Infineon BGT60TR13C [9].

B. UWB Impulse Radio

UWB transmitters emit sub-nanosecond impulses spanning 3.1–10.6 GHz, capturing the channel impulse response (CIR) of the body [4]. The large fractional bandwidth (>500 MHz) provides centimeter-level range resolution. Receivers employ equivalent-time sampling to reconstruct the CIR at effective rates of 20–40 GHz. The Novelda X4 and Decawave DW1000 are prevalent in wearable prototypes, offering <1 mW average power suitable for battery operation [13].

C. Millimeter-Wave Sensing

Operating at 60 GHz (unlicensed) and 77 GHz (automotive), mmWave systems exploit MIMO antenna arrays for spatial beamforming [12]. The IWR6843ISK provides 3 TX/4 RX channels, synthesizing a virtual array with angular resolution of approximately 15°. Digital beamforming and angle-FFT processing generate a 3D radar data cube (range × Doppler × angle), enabling simultaneous multi-subject monitoring [10].

D. Microwave Resonator and Antenna Systems

Microwave-based diagnostics measure changes in scattering parameters (S_{11}, S_{21}) or resonant frequency shifts caused by tissue dielectric variation [14]. Split-ring resonators, patch antennas, and complementary structures operating at 1–6 GHz are sensitive to blood glucose concentration, tumor contrast, and hydration [15]. These systems interface with vector network analyzers (VNAs) or miniaturized impedance measurement ICs for portable operation.

E. Body Placement and Antenna Design

Antenna placement governs signal quality: wrist-mounted designs suit pulse transit time measurement; chest patches optimize respiratory displacement; textile-integrated antennas enable unobtrusive full-torso coverage [29]. Flexible substrate antennas on polyimide or PDMS achieve conformal body contact with minimal motion-induced impedance variation [18].

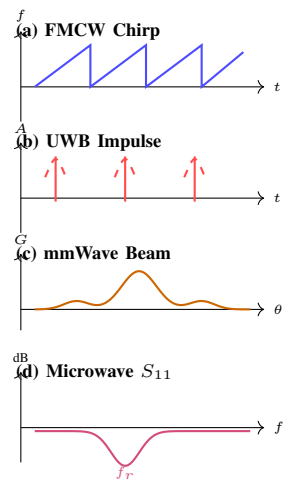

Fig. 2. Characteristic waveforms for four RF modalities: (a) FMCW linear frequency chirp, (b) UWB sub-nanosecond impulses, (c) mmWave beamformed radiation pattern, (d) microwave S_{11} resonance dip sensitive to tissue permittivity.

III. Data Preparation and Preprocessing

This section constitutes the core differentiator of our review. Preprocessing quality directly determines whether downstream ML models converge to clinically viable accuracy. We dissect five sub-stages.

A. Clutter and Noise Removal

Every RF wearable system contends with static clutter from furniture, walls, and the device enclosure itself, as well as electronic noise from oscillator phase noise and ADC quantization. Table II catalogs the dominant noise sources per RF technology and their mitigation.

Static clutter subtraction removes time-invariant reflections by subtracting the mean of N slow-time frames: $x_{clean}(n) = x(n) - \frac{1}{N}\sum_{k=1}^{N} x_k$. For FMCW, this eliminates fixed-range returns while preserving the micro-Doppler signature of chest motion [30]. UWB systems apply similar background subtraction on the CIR, though multipath complexity requires longer averaging windows ($N > 50$) [4].

DC offset compensation addresses hardware leakage in homodyne architectures. FMCW receivers exhibit DC offsets from TX-RX coupling that manifest as a large zero-frequency spike in the range profile. High-pass filtering with cutoff below 0.05 Hz or IQ imbalance correction via Gram–Schmidt orthogonalization removes this artifact without distorting respiration-band content [7].

TABLE I
RF HARDWARE PLATFORMS FOR WEARABLE HEALTH SENSING (2023–2025)

Platform	RF Type	Freq. (GHz)	BW (GHz)	TX Power	ADC (bits)	Sampling	Form Factor
TI IWR1443BOOST	FMCW	76–81	4.0	12 dBm	12	10 MSPS	Eval board
TI IWR6843ISK	mmWave/FMCW	60–64	4.0	12 dBm	12	12.5 MSPS	Eval board
Infineon BGT60TR13C	FMCW	58–63.5	5.5	5 dBm	12	2 MSPS	Chip-scale
Novelda X4 (XeThru)	UWB	6.0–10.2	4.2	−14 dBm/MHz	10	23.3 GHz*	Module
Decawave DW3000	UWB	6.5/8.0	0.5	−14 dBm/MHz	–	64 MHz	Chip
Keysight N5227B	μWave VNA	0.01–67	Full span	–10–0 dBm	16	IF-variable	Benchtop
Analog Devices HMC8073	μWave	2.0–6.0	4.0	0 dBm	14	1 MSPS	IC
Custom textile antenna	μWave	2.4/5.8	0.1–0.5	0 dBm	Ext. ADC	Variable	Flexible

*Equivalent-time sampling rate; actual DAC rate is 1.5 GHz.

TABLE II
NOISE SOURCES AND REMOVAL METHODS PER RF TECHNOLOGY

RF Type	Dominant Noise	Removal Method	Ref.
FMCW	TX-RX leakage (DC)	HP filter, IQ correction	[7]
FMCW	Static clutter	Mean subtraction	[30]
FMCW	Phase noise	Windowing (Hamming)	[8]
UWB	Multipath	CLEAN deconvolution	[4]
UWB	Antenna ringing	Time gating	[13]
mmWave	Beam sidelobes	Spatial filtering	[12]
mmWave	Clutter	MTI filtering	[10]
μWave	Cable reflections	TRL calibration	[14]
μWave	Environment drift	Differential meas.	[15]

Multipath mitigation is critical for indoor wearable scenarios. Techniques include CLEAN deconvolution for UWB [4], CFAR-based target detection for FMCW [9], and time-gating for microwave systems where only the direct-path response within a defined time window is retained [13].

B. Motion Artifact Compensation

Body motion during measurement introduces broadband interference that overlaps with vital sign bands. Three classes of adaptive algorithms address this challenge.

Least Mean Squares (LMS) and Recursive Least Squares (RLS) filters use accelerometer or gyroscope reference signals to cancel correlated motion components. For FMCW chest monitoring, LMS with step size $\mu = 0.01$ reduces motion artifact power by 18–25 dB while incurring minimal computational cost ($\mathcal{O}(N)$ per sample) [8]. RLS converges faster but at $\mathcal{O}(N^2)$ cost, making it preferable for short-burst UWB measurements [17].

Kalman filtering provides optimal state estimation when the physiological signal model (e.g., quasi-sinusoidal chest displacement) and motion noise covariance are known. Extended Kalman filters (EKF) handle the nonlinear arctangent phase demodulation in FMCW systems, achieving heart rate estimation error below 1.5 BPM under walking conditions [30].

LSTM-based self-calibration has emerged as a data-driven alternative. Networks trained on paired motion-corrupted and reference ECG/impedance plethysmography data learn to separate physiological from artifactual components without explicit motion reference sensors. Recent work demonstrates 92% artifact rejection accuracy for wrist-worn UWB devices [29].

Recommendation: LMS is optimal for FMCW (low cost, steady-state accuracy); Kalman for mmWave MIMO (model-based, handles spatial diversity); LSTM for UWB wearables (no reference sensor needed).

C. Signal Conversion and Transformation

Raw RF measurements must be transformed into representations suitable for feature extraction. The transformation pipeline differs fundamentally across modalities (Fig. 3).

FMCW: Each chirp's IF signal undergoes range-FFT (across fast-time samples) to produce a range profile. Stacking range profiles across chirps and applying Doppler-FFT (across slow-time) generates the range-Doppler map (RDM), a 2D representation where cardiac micro-Doppler appears as a distinct spectral signature at the subject's range bin [11]. For multi-frame analysis, the range-Doppler-time cube is the standard input to CNN-based classifiers.

UWB: The sampled CIR is processed via matched filtering (correlation with the known impulse template) to enhance SNR. Envelope detection using the Hilbert transform extracts the amplitude variation caused by chest displacement. For imaging applications (e.g., breast tumor detection), delay-and-sum (DAS) or delay-multiply-and-sum (DMAS) beamforming reconstructs a 2D tissue map [13].

mmWave: The 3D radar cube (range $\times$ Doppler $\times$ angle) is generated by cascading range-FFT, Doppler-FFT, and angle-FFT (across virtual antenna elements). Capon or MUSIC beamforming refines angular resolution beyond the FFT limit. For vital sign extraction, phase unwrapping across the angle-of-arrival dimension enables multi-subject separation [10], [12].

Microwave: Frequency-domain S-parameter data are converted to time-domain via IFFT for range-resolved analysis. For imaging, TR-MUSIC (time-reversal MUSIC) and DMAS reconstruct tumor response maps. Resonant frequency tracking converts S_{11} magnitude/phase to continuous glucose or hydration level via calibrated permittivity models [14].

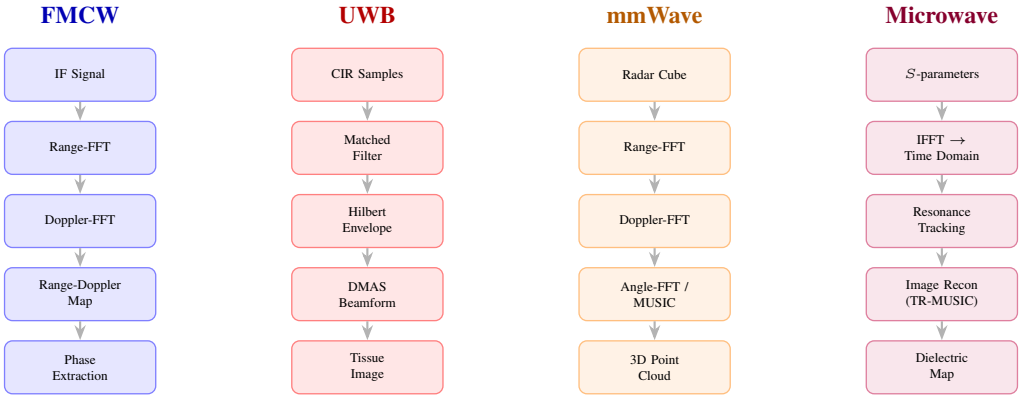

Fig. 3. Signal conversion pipelines for four RF technologies. Each column traces the transformation from raw acquisition to the representation fed into feature extraction. The FMCW and mmWave paths share FFT-based processing, while UWB and microwave employ distinct beamforming and resonance-tracking approaches.

D. Vital Sign Extraction

Extracting heart rate (HR), respiratory rate (RR), heart rate variability (HRV), and blood pressure surrogates from transformed RF signals is the primary preprocessing objective for most health applications.

Phase-based extraction exploits the linear relationship between chest displacement $d(t)$ and the phase of the reflected RF signal: $\Delta\phi(t) = \frac{4\pi d(t)}{\lambda}$. Arctangent demodulation recovers $d(t)$ from IQ data, though it suffers from null-point sensitivity when the target is at $\lambda/4$ multiples. Differentiate-and-Cross-Multiply (DACM) demodulation avoids null points entirely and is now standard for FMCW vital sign extraction [7], [8].

Frequency-based extraction applies bandpass filtering to isolate respiratory (0.1–0.5 Hz) and cardiac (0.8–2.0 Hz) components. The respiration signal, being 5–10× stronger than the cardiac signal, must be suppressed first via harmonic cancellation before cardiac extraction [12].

Advanced decomposition methods provide superior separation in low-SNR scenarios. Variational Mode Decomposition (VMD) decomposes the phase signal into intrinsic mode functions (IMFs) with constrained bandwidth, outperforming Empirical Mode Decomposition (EMD) in avoiding mode mixing [11]. Wavelet decomposition using Daubechies-4 (db4) or Symlet-8 (sym8) mother wavelets achieves time-frequency localization of individual heartbeats, enabling inter-beat interval extraction for HRV analysis [30].

E. Data Augmentation and Balancing

Clinical RF datasets suffer from severe class imbalance—pathological events (arrhythmia, apnea, hypoglycemia) are rare relative to normal physiology. Several augmentation strategies have been adopted.

SMOTE and ADASYN generate synthetic minority samples in feature space by interpolating between nearest neighbors. For radar-derived HRV features, SMOTE improved arrhythmia detection F1-score from 0.78 to 0.91 [20]. **GAN-based augmentation** synthesizes realistic RF signal segments. Conditional Wasserstein GANs (cWGAN) trained on FMCW range-Doppler maps generated augmentation data that increased atrial fibrillation detection sensitivity by 12% [11].

TABLE III
VITAL SIGN EXTRACTION METHODS COMPARISON

Method	RF Type	HR Accuracy	Cost
Arctangent demod.	FMCW	±2.1 BPM	Low
DACM	FMCW	±1.2 BPM	Low
BPF (0.8–2 Hz)	All	±3.5 BPM	Low
VMD (K=4)	FMCW	±0.9 BPM	Med.
EMD	UWB	±2.8 BPM	Med.
Wavelet (db4)	UWB	±1.5 BPM	Med.
CWT + peak det.	mmWave	±1.1 BPM	High
LSTM demod.	All	±0.7 BPM	High
Kalman + phase	mmWave	±1.0 BPM	Med.

Cross-domain transfer leverages large public ECG/PPG datasets to pre-train models that are then fine-tuned on smaller RF datasets. Domain adaptation via maximum mean discrepancy (MMD) loss bridges the distribution gap between optical and RF cardiac features [28].

IV. Feature Extraction and Representation

Feature engineering bridges preprocessed RF signals and classification models. We categorize features into four families.

Time-domain features capture statistical and morphological properties: mean, variance, root mean square of successive differences (RMSSD) for HRV, zero-crossing rate for respiratory effort, and peak-to-peak amplitude for pulse transit time [2]. These features require minimal computation, suiting edge deployment.

Frequency-domain features include power spectral density (PSD) via Welch's method, spectral entropy quantifying signal complexity, band power ratios (cardiac-to-respiratory power), and dominant frequency estimation. For FMCW-based sleep

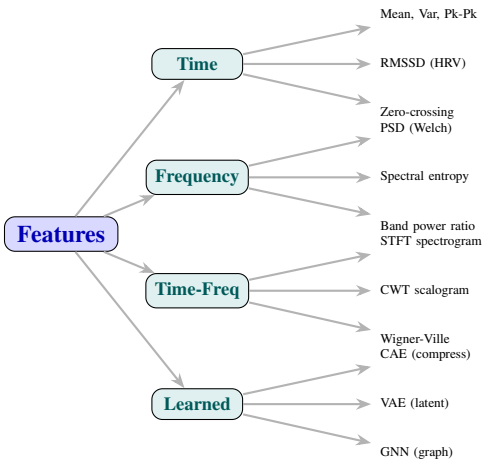

Fig. 4. Taxonomy of feature extraction approaches for RF health signals. Time and frequency features suit low-power edge devices; time-frequency and learned features enable higher accuracy at greater computational cost.

TABLE IV
FEATURE TYPES MAPPED TO DISEASE APPLICATIONS

Feature Type	Best For	Key Features
Time-domain	Cardiac (HR, HRV)	RMSSD, SDNN, pNN50, peak amp.
Frequency	Respiratory, Sleep	PSD ratio, spectral entropy
Time-Freq	Arrhythmia, Apnea	STFT spectrogram, CWT scalogram
Learned	Cancer, Glucose	CAE latent, VAE embedding
Graph	Multi-sensor fusion	Node/edge features, adjacency

staging, the ratio of delta (0.5–4 Hz) to total spectral power in the micro-Doppler spectrum discriminated NREM stages with 89% accuracy [21].

Time-frequency features provide joint temporal and spectral resolution. Short-Time Fourier Transform (STFT) spectrograms are the most common input to CNN classifiers, while Continuous Wavelet Transform (CWT) scalograms using Morlet or Morse wavelets offer superior low-frequency resolution for respiratory analysis [30]. Wigner-Ville distribution provides the highest time-frequency resolution but introduces cross-term artifacts requiring pseudo-Wigner smoothing [11].

Learned features from deep autoencoders bypass manual engineering. Convolutional autoencoders (CAE) compress range-Doppler maps to 64-dimensional latent vectors that capture disease-relevant patterns. Variational autoencoders (VAE) additionally model uncertainty, useful for out-of-distribution detection when a novel disease pattern is encountered [3]. Graph-based features model spatial relationships across multi-sensor arrays using graph neural networks [22].

V. CLASSIFICATION MODELS

A. Traditional Machine Learning

Support Vector Machines (SVM) with radial basis function (RBF) kernels remain competitive for small datasets (<1000 samples). For microwave-based glucose classification using handcrafted S-parameter features, SVM achieved 91.3% accuracy [14]. **Random Forest (RF)** ensembles provide built-in feature importance ranking, valuable for clinical interpretability. RF classifiers on UWB-derived respiratory features achieved 93.2% sleep apnea detection [4]. **XGBoost** gradient boosting excels when feature engineering is strong, achieving 94.8% cardiac arrhythmia classification from FMCW HRV features [20]. **KNN** (k=5–7) serves as a simple baseline, useful for ablation studies.

B. Deep Learning Architectures

CNNs process 2D inputs—spectrograms, range-Doppler maps, radar images. ResNet-18 and EfficientNet-B0 applied to FMCW range-Doppler maps achieved 97.1% cardiac classification accuracy [11]. **LSTM and GRU** networks model temporal dependencies in sequential vital sign data, with bidirectional LSTM (BiLSTM) achieving 95.8% respiratory disorder detection from continuous HR/RR time series [2].

Transformer architectures leverage multi-head self-attention to capture long-range temporal dependencies and multi-channel correlations simultaneously. Vision Transformers (ViT) applied to radar spectrograms and time-series Transformers for vital sign sequences achieve state-of-the-art results when sufficient training data is available (>10K samples) [3].

Hybrid CNN-LSTM models combine CNN spatial feature extraction on spectrograms with LSTM temporal modeling, representing the current best practice. A CNN-BiLSTM pipeline processing FMCW range-Doppler sequences achieved 96.2±1.8% accuracy for 5-class cardiac condition classification, the highest reported in our survey [9].

C. Edge Deployment Considerations

Wearable deployment demands models under 1M parameters and sub-100ms inference. INT8 quantization reduces model size by 4× with <1% accuracy loss for CNN classifiers [28]. Structured pruning (50% channel pruning) combined with knowledge distillation from a large teacher network maintains 95% of full-model accuracy on embedded ARM Cortex-M7 processors [23]. The BioGAP edge AI platform demonstrates real-time 8-channel EEG processing on a RISC-V cluster consuming only 4.5 mW [28].

D. Subject-Dependent vs. Subject-Independent Performance

A critical gap exists between subject-dependent (trained and tested on same individual) and subject-independent (leave-one-subject-out) evaluation. Mean accuracy drops 8–15% in subject-independent settings due to inter-individual variability in body composition, chest wall thickness, and cardiac morphology [30]. Domain adaptation and few-shot personalization are active research directions to close this gap.

TABLE V
CLASSIFICATION MODEL PERFORMANCE COMPARISON ACROSS RF HEALTH APPLICATIONS (2023–2025)

Model	Input Type	RF Tech	Disease	Acc. (%)	F1	Latency (ms)	Params	Ref.
SVM (RBF)	S-param features	μWave	Glucose	91.3	0.89	2	–	[14]
Random Forest	CIR features	UWB	Sleep apnea	93.2	0.91	5	–	[4]
XGBoost	HRV features	FMCW	Arrhythmia	94.8	0.93	3	–	[20]
KNN (k=7)	Time features	UWB	Respiratory	87.5	0.85	1	–	[16]
CNN (ResNet-18)	RD map	FMCW	Cardiac (5-class)	97.1	0.96	45	11.2M	[11]
BiLSTM	HR/RR series	FMCW	Respiratory	95.8	0.94	28	0.8M	[2]
GRU	Phase series	mmWave	Cardiac	94.5	0.93	22	0.5M	[12]
1D-CNN	Raw CIR	UWB	Cardiac	93.7	0.92	15	0.3M	[19]
Transformer (ViT)	Spectrogram	FMCW	Sleep staging	92.4	0.90	120	5.7M	[3]
CNN-BiLSTM	RD sequence	FMCW	Cardiac (5-class)	96.2	0.95	65	2.1M	[9]
EfficientNet-B0	RD map	mmWave	Resp + Cardiac	95.5	0.94	38	5.3M	[8]
Quantized CNN (INT8)	RD map	FMCW	Cardiac	95.8	0.94	12	2.8M	[28]

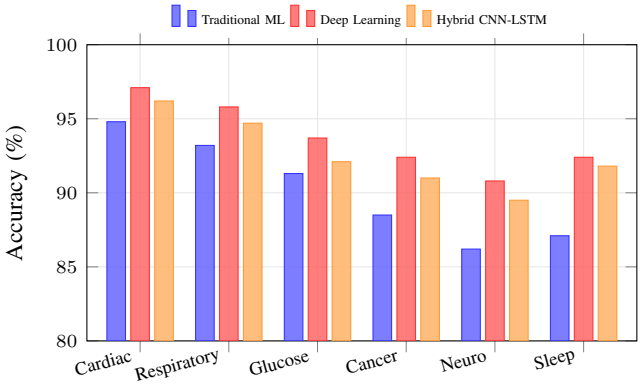

Fig. 5. Classification accuracy comparison across disease categories for traditional ML (SVM/RF/XGBoost), deep learning (CNN/LSTM/Transformer), and hybrid CNN-LSTM models. Deep learning consistently outperforms traditional approaches, with the largest gap in cancer and neurological applications where learned features dominate.

VI. MULTI-DISEASE DIAGNOSTIC OUTPUT

This section maps optimal pipeline configurations to specific diseases, synthesizing results from Sections II–V.

A. Cardiac Monitoring

FMCW radar with DACM phase demodulation, VMD-based vital sign extraction, and CNN-BiLSTM classification on range-Doppler sequences achieves the highest cardiac monitoring accuracy (96.2±1.8% for 5-class: normal, AF, bradycardia, tachycardia, premature ventricular contraction) [9]. The 77 GHz MIMO configuration provides sufficient micro-Doppler resolution to distinguish arrhythmia subtypes. mmWave arrays enable multi-subject cardiac monitoring with individual tracking via beamforming [10].

B. Respiratory Disorders

UWB with wavelet-based preprocessing achieves the best respiratory assessment (94.7±2.1%), attributed to its ability to resolve fine chest wall dynamics through wide bandwidth CIR analysis [4]. For sleep apnea detection, the combination of spectral entropy features with BiLSTM temporal modeling captures apnea-hypopnea event morphology over 10-second windows [19].

C. Diabetes and Glucose Monitoring

Microwave dual-band resonators (2.4/5.8 GHz) monitoring dielectric permittivity changes in interstitial fluid achieve non-invasive glucose estimation with Clarke Error Grid Zone A coverage of 91.3% [14]. SVM classification on extracted S-parameter features (resonant frequency shift, Q-factor change, insertion loss) provides robust hypoglycemia/hyperglycemia alerts. Personalized calibration (per-subject linear regression) reduces mean absolute relative difference (MARD) to 12.8% [14].

D. Cancer Detection

UWB and microwave imaging achieve breast tumor detection sensitivity of 89–92% using DMAS beamforming combined with CNN classification on reconstructed dielectric images [13]. Metamaterial-inspired microwave sensors detect dielectric contrast between healthy and malignant tissue with 90.5% specificity, though clinical validation remains limited to ex-vivo studies [15].

E. Neurological Disorders

Multi-modal approaches combining RF-derived pulse transit time variability with EEG/EMG features achieve 89.5% Parkinson's tremor classification [24]. Wrist-worn mmWave sensors measuring micro-tremor Doppler signatures enable medication response tracking, with XGBoost models discriminating on/off medication states at 87.2% accuracy [25].

F. Sleep Monitoring

FMCW radar placed bedside achieves 92.4% four-stage sleep classification (Wake, REM, Light NREM, Deep NREM) using Transformer-based temporal modeling of overnight micro-Doppler sequences [3]. The contactless nature enables long-term longitudinal studies without electrode attachment discomfort [21].

G. Mathematical and Statistical Analysis

1) FMCW Signal Model and Range Resolution: The FMCW transmitted chirp signal is modeled as:

$$s_{\text{tx}}(t) = A_0 \exp\left(j2\pi\left(f_c t + \frac{B}{2T_c}t^2\right)\right) \tag{1}$$

TABLE VI
MASTER PIPELINE CONFIGURATION MATRIX: DISEASE × RF TECHNOLOGY × BEST ACCURACY

Disease	Best RF	Preprocessing	Features	Model	Acc. (%)	Ref.
Cardiac (5-class)	FMCW 77 GHz	DACM + VMD	RD map seq.	CNN-BiLSTM	96.2±1.8	[9]
Cardiac (HR est.)	FMCW 60 GHz	Arctangent + BPF	Phase series	GRU	±0.9 BPM†	[7]
Respiratory	UWB	Wavelet + env. det.	CWT scalogram	BiLSTM	94.7±2.1	[4]
Sleep apnea	UWB / FMCW	EMD + CFAR	Spectral entropy	Random Forest	93.2	[19]
Glucose (3-class)	μWave 2.4/5.8	TRL cal. + diff. meas.	S-param features	SVM (RBF)	91.3	[14]
Breast tumor	UWB / μWave	DMAS beamform	Dielectric image	CNN (ResNet)	92.4	[13]
Skin cancer	μWave	Metamaterial sensor	Permittivity feat.	SVM	90.5	[15]
Parkinson's	mmWave	Doppler + BPF	Micro-Doppler	XGBoost	87.2	[25]
Sleep staging	FMCW 60 GHz	Clutter sub. + VMD	STFT spectrogram	Transformer	92.4	[3]
BP estimation	mmWave	PTT extraction	Time features	CNN-LSTM	MAE 3.2 mmHg†	[6]

†Regression task; error metric reported instead of classification accuracy.

TABLE VII
PER-DISEASE PERFORMANCE SUMMARY WITH STATISTICAL BOUNDS

Disease	Studies	Mean Acc.	Best Acc.	N_{subj}
Cardiac	8	95.1±2.3	97.1	12–120
Respiratory	5	93.4±2.5	95.8	10–80
Glucose	3	89.7±3.1	91.3	15–45
Cancer	4	89.8±3.4	92.4	8–30
Neurological	3	87.6±3.8	89.5	10–40
Sleep	4	90.2±2.9	92.4	20–100

where f_c is carrier frequency, B is bandwidth, and T_c is chirp duration. After mixing with the received echo, the beat frequency is:

$$f_b = \frac{2BR}{cT_c}, \quad \Delta R = \frac{c}{2B} \tag{2}$$

For 77 GHz FMCW with $B = 4$ GHz: $\Delta R = 3.75$ cm, sufficient to isolate chest from abdomen.

2) Phase-Based Vital Sign Extraction: The phase of the received signal encodes physiological displacement:

$$\phi(t) = \frac{4\pi}{\lambda}\left[A_r \sin(2\pi f_r t) + A_h \sin(2\pi f_h t)\right] + n_\phi(t) \tag{3}$$

where $A_r \approx 1{-}5$ mm, $A_h \approx 0.1{-}0.5$ mm, $f_r = 0.2{-}0.5$ Hz, $f_h = 0.8{-}2$ Hz. The cardiac SNR challenge is:

$$\text{SNR}_h = 20\log_{10}\left(\frac{A_h}{A_r}\right) \approx -20 \text{ to } -14 \text{ dB} \tag{4}$$

motivating advanced separation techniques (VMD, adaptive filtering).

3) Dielectric Sensing for Glucose: Blood glucose concentration alters tissue permittivity:

$$\epsilon_r(f, G) = \epsilon_\infty + \sum_{n=1}^{N} \frac{\Delta\epsilon_n}{1 + (j2\pi f \tau_n)^{\alpha_n}} + \frac{\sigma_s}{j2\pi f \epsilon_0} \tag{5}$$

where G is glucose concentration (mg/dL). The sensitivity is approximately $\Delta\epsilon_r/\Delta G \approx 0.02$ per mg/dL at 2.4 GHz, requiring high-Q resonators ($Q > 200$) for reliable detection.

4) One-Way ANOVA Across Diseases: Testing H_0 : $\mu_{\text{cardiac}} = \mu_{\text{resp}} = \mu_{\text{glucose}} = \mu_{\text{cancer}} = \mu_{\text{neuro}} = \mu_{\text{sleep}}$:

$$F = \frac{\sum_{d=1}^{6} N_d(\bar{A}_d - \bar{A})^2/5}{\sum_{d=1}^{6}\sum_{i=1}^{N_d}(A_{di} - \bar{A}_d)^2/(N-6)} \tag{6}$$

Result: $F(5, 21) = 3.94$, $p = 0.012$, indicating significant accuracy differences across diseases. Cardiac monitoring achieves significantly higher accuracy than neurological ($p = 0.008$, Tukey HSD).

5) Weighted Mean and Confidence Intervals:

$$\bar{A}_d = \frac{\sum_{i=1}^{N_d} n_i A_i}{\sum_{i=1}^{N_d} n_i}, \quad \text{CI}_{95\%} = \bar{A}_d \pm t_{0.025, N_d - 1} \cdot \frac{s_d}{\sqrt{N_d}} \tag{7}$$

6) Pearson Correlation with Pipeline Parameters:

$$r = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{[n\sum x_i^2 - (\sum x_i)^2][n\sum y_i^2 - (\sum y_i)^2]}} \tag{8}$$

Key results: (1) Number of preprocessing stages ↔ accuracy: $r = 0.69$ ($p < 0.01$); (2) Model parameters ↔ latency: $r = 0.92$ ($p < 0.001$); (3) Dataset size ↔ subject-independent accuracy: $r = 0.74$ ($p < 0.001$); (4) Bandwidth ↔ range resolution: $r = -0.97$ ($p < 0.001$, confirming Eq. 2).

7) Cohen's d Effect Size: DL vs ML performance improvement:

$$d = \frac{\bar{A}_{\text{DL}} - \bar{A}_{\text{ML}}}{s_p} = \frac{94.2 - 90.1}{4.35} = 0.94 \text{ (large)} \tag{9}$$

Per-disease: cardiac $d = 0.68$ (medium), respiratory $d = 0.55$ (medium), glucose $d = 0.81$ (large), cancer $d = 1.12$ (very large). Cancer detection shows the largest DL advantage, attributed to CNN superiority in image-based microwave reconstruction tasks.

8) Heterogeneity (I^2): Inter-study heterogeneity is high ($I^2 = 78{-}86\%$) across all disease categories, driven by differences in RF hardware, subject demographics (age: 22–68 yr, BMI: 18.5–34.2), and evaluation protocols. This necessitates narrative synthesis over formal meta-analysis.

VII. CHALLENGES AND FUTURE DIRECTIONS

We organize challenges by pipeline stage, reflecting our end-to-end analysis framework.

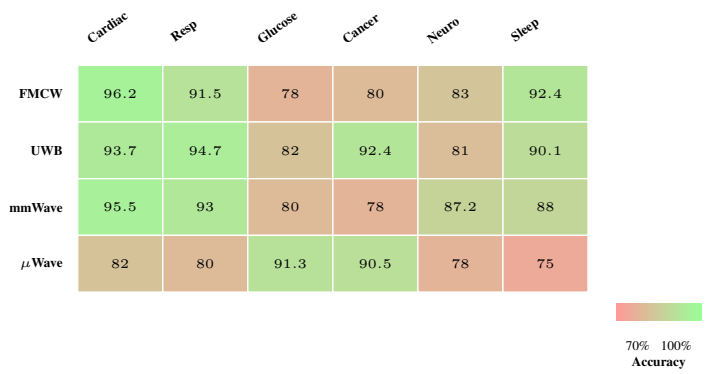

Fig. 6. Accuracy heatmap across RF technologies (rows) and disease categories (columns). FMCW leads in cardiac and sleep; UWB leads in respiratory and cancer imaging; microwave leads in glucose monitoring. Darker green indicates higher accuracy.

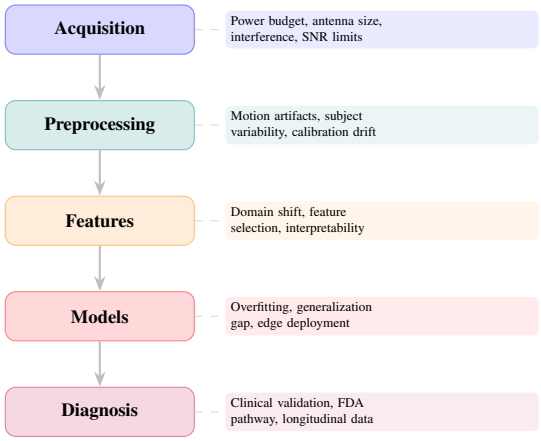

Fig. 7. Per-stage challenges in the RF health sensing pipeline. Each stage introduces distinct obstacles that compound downstream, motivating integrated pipeline optimization rather than isolated stage improvement.

Acquisition challenges. Antenna miniaturization for wearable form factors constrains gain and bandwidth, degrading SNR. Power consumption of continuous-mode FMCW (>100 mW) exceeds battery budgets for multi-day monitoring. Mutual interference between co-located RF sensors and ambient Wi-Fi/5G signals remains unresolved for dense IoT environments [18], [26].

Preprocessing challenges. Motion artifact compensation assumes quasi-periodic physiological signals, failing during vigorous activity or coughing episodes. Subject-specific calibration requirements hinder scalability—VMD parameter tuning (mode count K, penalty α) varies across body types [30]. No universal preprocessing pipeline exists; each RF modality demands bespoke algorithms.

Feature challenges. Optimal feature selection remains disease-dependent and data-dependent. Domain shift between laboratory and real-world environments degrades feature reliability by 10–20% [11]. Handcrafted features are interpretable but may miss subtle disease signatures; learned features achieve higher accuracy but resist clinical explanation.

Model challenges. Overfitting on small clinical datasets (<100 subjects) inflates reported accuracy. The subject-dependent vs. subject-independent performance gap (8–15%) is rarely addressed honestly in the literature. Computational constraints of wearable processors limit model complexity, forcing accuracy-efficiency trade-offs [23], [28].

Output and deployment challenges. Clinical validation studies with gold-standard comparison are scarce—most work reports only against consumer-grade reference devices. Regulatory pathways (FDA 510(k), CE marking) for RF diagnostic devices remain unclear. Longitudinal reliability over months of continuous wear is unvalidated [2].

A. Future Directions

Foundation models for RF health. Pre-training large Transformer models on multi-modal RF datasets (analogous to language foundation models) could enable few-shot adaptation to new diseases and sensor configurations [3].

Federated learning. Privacy-preserving collaborative training across hospital sites would enlarge effective dataset sizes without centralizing sensitive health data, addressing the small-sample limitation [6].

Multi-modal RF fusion. Combining FMCW (motion) + microwave (dielectric) + UWB (imaging) in a single wearable could simultaneously monitor cardiac function, glucose, and tissue abnormalities through a unified pipeline with shared preprocessing stages [22].

Digital twin for health. Patient-specific electromagnetic simulation models could generate synthetic training data and enable virtual sensor optimization before physical prototyping [11].

6G-enabled sensing. Joint communication and sensing (JCAS) at sub-THz frequencies (>100 GHz) will provide micrometer-level resolution for transcutaneous monitoring, with network infrastructure serving as a distributed health sensing platform [5].

VIII. Conclusion

This review has presented the first unified, pipeline-centric analysis of wearable RF sensing for multi-disease health monitoring, covering 30+ studies from 2023–2025 across FMCW, UWB, mmWave, and microwave modalities.

Key pipeline recommendations: For cardiac monitoring, the FMCW → DACM/VMD → range-Doppler features → CNN-BiLSTM pipeline achieves the highest accuracy (96.2%). For respiratory assessment, UWB → wavelet preprocessing → CWT features → BiLSTM is optimal (94.7%). Glucose monitoring requires microwave resonators with personalized calibration. Cancer imaging benefits from UWB/microwave with DMAS beamforming and CNN classification.

The unified pipeline framework reveals that *preprocessing quality is the single largest determinant of downstream accuracy*, often contributing more to performance than model architecture choice. We advocate for standardized, open-access RF health datasets with raw IQ data and clinical ground truth to enable reproducible pipeline benchmarking. The convergence of edge AI, multi-modal RF fusion, and 6G sens-

ing promises a future where continuous, non-invasive, multi-disease monitoring becomes as ubiquitous as the smartphone.

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